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SARDAR PATEL UNIVERSITY
External Examination
BCA(NEP) SEMESTER-III
Discrete Mathematics (US03AEBCA06)



16-2-2026
Monday

TIME: 02:00PM to 03:30 PM
TOTAL MARKS: 25

Notes

1. Figures to the right indicate full marks
2. All questions are compulsory

Q-1 Multiple Choice Questions

[05]

(1) If A is a 2×3 matrix and B is a 3×2 matrix, what is the order of the product AB?

- (A) 2×2 (B) 3×3 (C) 2×3 (D) 3×2

(2) Which type of matrix has the property that $A = A^T$?

- (A) Skew-Symmetric (B) Diagonal (C) Symmetric (D) Identity

(3) What is transpose of a scalar matrix A?

- (A) A (B) -A (C) Zero Matrix (D) Undefined

(4) The Euler's formula is _____

- (A) $R + E - V = 2$ (B) $V - E - R = 2$ (C) $E - R + V = 2$ (D) $V - E + R = 2$

(5) What is the chromatic number of a complete graph K_n ?

- (A) 1 (B) n (C) n-1 (D) 2

Q-2

[A] If $A = \begin{bmatrix} 4 & -1 \\ -1 & 3 \\ 2 & 0 \end{bmatrix}$, $B = \begin{bmatrix} -2 & 5 \\ 3 & -1 \\ 5 & 2 \end{bmatrix}$ and $C = \begin{bmatrix} 2 & 4 \\ -1 & -5 \\ 3 & -2 \end{bmatrix}$,

[10]

Then find 1. $A + B + C$ 2. $3A - 4B + C$

[B] Let $\vec{u} = (1, 2, 3, -4)$, $\vec{v} = (5, -6, 7, 8)$ and $k = 3$.

Find i) $k(\vec{u} \cdot \vec{v})$ ii) $(k\vec{u}) \cdot \vec{v}$ iii) $\|\vec{u} + \vec{v}\|$ iv) $\|3\vec{u}\|$

OR

Q-2

[A] If $A = \begin{bmatrix} 1 & -2 & 3 \\ 2 & 3 & -1 \\ -3 & 1 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 2 \\ 1 & 2 & 0 \end{bmatrix}$, find AB and BA. Show

[10]

that $AB \neq BA$.

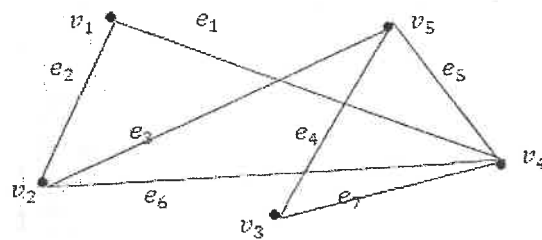
[B] If $A = \begin{bmatrix} -1 & 2 & 6 \\ 6 & 2 & -1 \\ 3 & 4 & 7 \end{bmatrix}$ $B = \begin{bmatrix} 3 & 4 & 5 \\ 4 & 3 & 2 \\ 0 & 2 & -1 \end{bmatrix}$ Find $A^2 + B$.

①

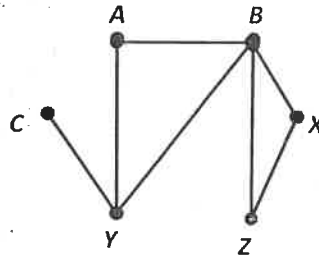
(P.T.O.)

Q-3 [A] Find the adjacency & Incidence matrix of the graph given below;

[10]



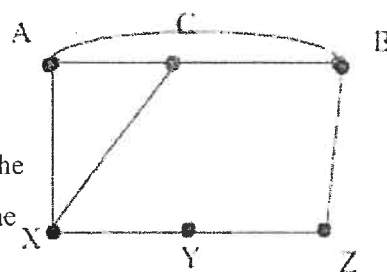
[B] What is a cut point? Find all cut points in the following graph.



OR

Q-3 [A] Consider the graph:->

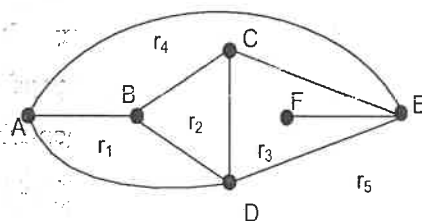
- (1) Describe the graph formally.
- (2) Find the degree of each vertex.
- (3) Verify that the sum of the degrees of the Vertices of a graph is equal to twice the Number of edges.



[10]

[B] Consider the graph given below:

- (1) State Euler's formula for the map. Verify it for the map given
- (2) Define region, degree of region, find it for the map given. Also Verify the theorem $\deg(r_i) = 2(e_i)$ & colour the map.



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